Ernest Ma

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Education

San Jose State University **December 2026**

Bachelor of Science in Computer Engineering | GPA: 3.95 *San Jose, CA*

- **Organizations**: Software and Computer Engineering Society, Spartan Racing: Formula SAE, HKSA
- **Relevant Coursework**: Data Structures and Algorithms, RTOS Embedded Systems, Microprocessor Design, Operating Systems, Database Systems, Mobile Software Engineering, Computer Networks, Digital Design

Experience

KLA | Software Engineer Intern Ann Arbor, MI | May 2025 – August 2025

- Developed a troubleshooting agent using **LangGraph**, **FastAPI**, and **Gemma 3 / GPT-4o** to assist field engineers.
- Reduced machinery downtime by **80%** by automating parsing of complex system logs and enabling natural language queries.
- Engineered a **RAG** pipeline with ChromaDB to accelerate historical log retrieval from hours to sub-second results.

SJSU College of Engineering | Software Engineer Intern San Jose, CA | January 2026 – Present

- Architected a **Go/Gin** event service processing registrations and waitlists for **1k+** concurrent users via a MongoDB backend
- Mitigated data races during **10k+ write/sec** spikes by implementing **Redis** concurrency locks and optimized caching strategies
- Orchestrated a **Kafka** event-driven pipeline to decouple API requests from registration processing, ensuring strong scalability

Spartan Racing: Formula SAE | Software Engineer Intern San Jose, CA | January 2026 – Present

- Expanded a **C-based** Hardware in Loop injection system for testing using CAN protocols for real-time VCU validation
- Developed a **Python** HIL test suite utilizing state machines to manage complex CAN communication sessions
- Synchronized automated telemetry validation to ensure data consistency during high-speed testing cycles
- Verified Launch Control accuracy by hand-calculating torque outputs against parameterized test cases

KLA | Student Ambassador San Jose, CA | January 2026 – Present

- Represented and spread awareness of KLA at campus recruiting / club events, info sessions, and career fairs
- Coordinated with KLA recruiters to organize networking events to increase student participation with **100+ attendees**

Projects

SJSU Safeline — *Nemotron, FastAPI, React / Vite, LangChain, SQLite*

- Built an agentic AI assistant using **NVIDIA Nemotron** and FastAPI to automate crisis triaging and personalized resource routing
- Architected a reasoning pipeline with LangChain function-calling for urgency assessment and real-time resource retrieval
- Implemented a multi-stage workflow to provide 24/7 student support through generating personalized outreach scripts

Fynd — *Flutter (Dart), Firebase, NoSQL*

- Developed a full-stack Flutter app using Firebase Auth and Firestore for real-time partner linking and data synchronization
- Engineered an on-device recommendation engine wth a Contextual Bandit model with Thompson Sampling for minimal latency
- Built a hybrid scoring pipeline using Cosine and Jaccard Similarity to generate thematic gift recommendations from user likes
- Implemented Dart-based matrix utilities for Cholesky decomposition and diversity filtering to ensure high-relevance results

Collab Code — *TypeScript, VS Code Extension API, Liveblocks, Yjs*

- Built a real-time collaborative IDE extension using Yjs CRDTs and Liveblocks for multi-user editing and room sessions
- Implemented synchronized cursors, selections, and presence indicators with document binding and local-origin filtering
- Created account-less shared-key onboarding and automated VSIX packaging for seamless cross-editor distribution

Pokewatcher — *Python, FastAPI, BeautifulSoup4, APScheduler, Discord API*

- Programmed an autonomous eBay Pokemon Card watcher for FastAPI system and Discord alerts API
- Architected a pipeline processing 10,000+ daily listings, utilizing BeautifulSoup4 to scrap for pricing
- Enhanced system performance and security by implementing Fernet-encrypted credential storage and automated data lifecycle jobs, reducing database storage overhead by 40% through stale telemetry cleanup.

Technical Skills

Languages: Go, Python, SQL, C++, Verilog, Dart, x86 Assembly, C, C#, Java
Tools and Platforms: Git, Docker, MongoDB, Redis, Kafka, Firebase, Vercel, Fly, Vivado, Quartus, ModelSim, Linux